

Exploratory Data Analysis: Executive Summary

Project 2 | DecodeLabs Industrial Training Kit, Batch 2026 | Prepared by Sajeel Ahmed

This dataset is synthetic. It is not suitable for business decisions.

Dataset 1,200 rows, 14 columns	Period covered Jan 2023 to Jun 2025	Tools Python, pandas, matplotlib
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1. Problem Statement

Can this dataset be used for business decisions? Project 1 confirmed the dataset is structurally clean, with no duplicates and no format errors. Structural cleanliness, however, says nothing about whether the values themselves are meaningful. This analysis examines the distributions, relationships and trends inside the data to determine whether any of it can support a business conclusion.

2. Methodology

Five techniques were applied to the four numeric columns, using Python and pandas:

- Descriptive statistics and the five number summary, to locate the centre and spread of each column.
- Distribution shape, measured with a skew coefficient and confirmed visually with histograms.
- Outlier detection using the IQR method, with bounds set at 1.5 times the interquartile range.
- Correlation analysis across all numeric pairs.
- Monthly revenue aggregation across the 30 month period, to test for trend or seasonality.

3. Key Findings

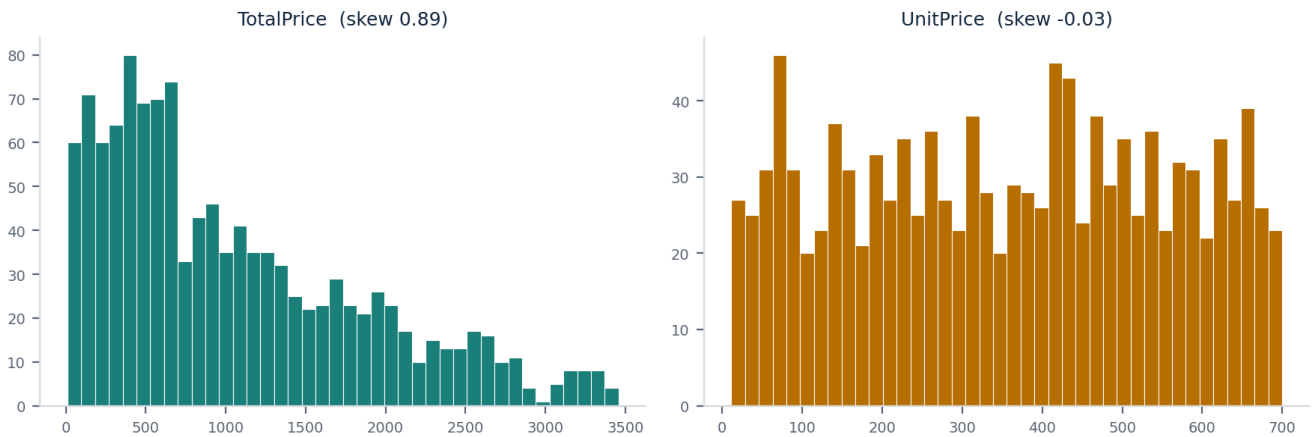
ID	Observation	Interpretation
F1	UnitPrice is uniformly distributed (skew of -0.03). Prices are spread evenly from 11.39 to 699.93 with no peak anywhere in the range.	Real pricing never behaves this way. Genuine sales data clusters around common price points. This is the strongest single piece of evidence that the values were generated at random.
F2	Quantity (skew 0.028) and ItemsInCart (skew 0.0009) are almost perfectly symmetrical.	Three of the four numeric columns are flat. Real transaction data always carries some natural imbalance.
F3	TotalPrice is the only skewed column (skew 0.89), but the skew is mathematical rather than behavioural. TotalPrice equals Quantity multiplied by UnitPrice, and multiplying two symmetrical values always produces a right skew.	Reporting this as evidence that some customers spend far more would be a false conclusion. The pattern comes from the formula, not from the customers.
F4	Only one genuine correlation exists: Quantity and ItemsInCart at 0.650. The correlations between TotalPrice and its own components (0.717 and 0.615) are structural and carry no information.	ItemsInCart is never lower than Quantity in any of the 1,200 rows. A relationship that holds in 100 percent of cases points to a generation rule rather than customer behaviour.
F5	Monthly revenue moves between roughly 28,000 and 68,000 with no direction and no repeating seasonal pattern.	This is noise, not a trend. With around 40 orders per month, a single large order visibly moves the monthly total.

Outliers

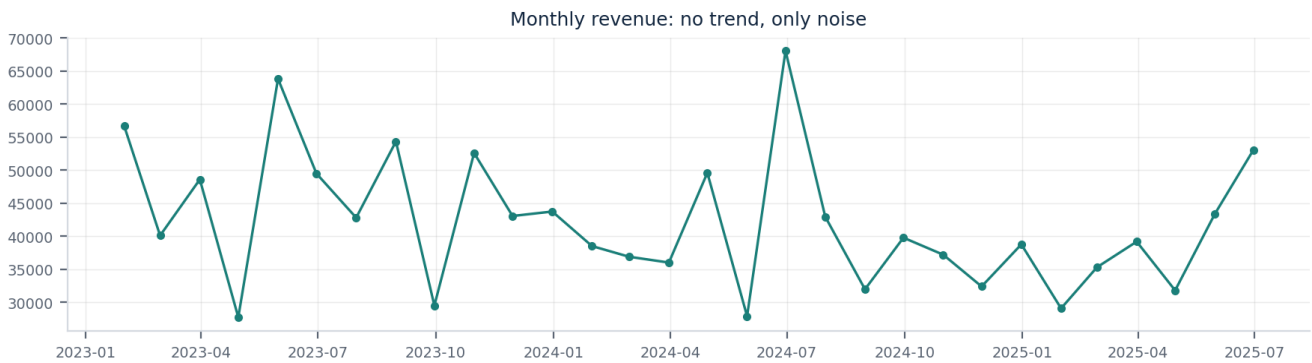
The IQR method flagged 8 orders above 3,330.41. All eight combine the maximum quantity of 5 with a unit price near the maximum of 699.93. Every value sits inside the valid range of the dataset, so none of them are data entry errors. They are statistical outliers rather than business outliers, and they were retained. The lower IQR bound calculated to

-1,341.41, a threshold no order value could ever fall below, so no outliers were possible on that side.

Visual Evidence



Left: TotalPrice tapers into a long right tail. Right: UnitPrice is flat across its entire range, which is the signature of randomly generated values.



Monthly revenue reverses direction almost every month, with no sustained movement in either direction and no seasonal repetition across the two and a half year period.

4. Recommendations

Do not use for	Reason
Pricing decisions	UnitPrice carries no relationship to product, so any price conclusion would be arbitrary.
Revenue forecasting	No trend or seasonality exists in the 30 month series. There is nothing to project forward.
Customer segmentation	Spending differences reflect the multiplication of two random values, not customer behaviour.
Geographic analysis	All 1,200 addresses share a single street name, with no city, state or postcode.

The dataset remains suitable for technical practice, which is what it appears designed for. Cleaning, statistical technique and visualisation can all be learned on it. Any conclusion about the business it claims to describe would be invented rather than discovered. Genuine transaction data is required before this type of analysis can inform a decision.